

ON THE CHERN-RICCI FORM OF A TWISTED ALMOST KÄHLER STRUCTURE

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ABSTRACT. Let (M, g, J, ω) be an almost Kähler manifold. For any smooth function f on M , one can associate an automorphism $\psi \in \text{Aut}(TM)$ for which the Kähler form is invariant. Using ψ , one can “twist” the metric g and almost complex structure J to obtain a new almost Kähler structure (g^ψ, J^ψ, ω) on M . Let $\tilde{D}$ denote the Chern connection of (g^ψ, J^ψ, ω) and let K^{-1} denote the anti-canonical bundle of (TM, J^ψ) . In the current paper, we give an explicit formula for the local connection 1-form α associated to the pair $(K^{-1}, \tilde{D})$. The Chern-Ricci form of (g^ψ, J^ψ, ω) is then $\rho_{\tilde{D}} = -d\alpha$. We note that under certain conditions the aforementioned formula assumes a simpler form when applied to the calculation of α . We illustrate this with some examples.

1. INTRODUCTION

Let (M, g, J, ω) be an almost Kähler manifold where

- g is the metric,
- J is the almost complex structure, and
- $\omega := g(J\cdot, \cdot)$ is the Kähler form.

In other words, (g, J, ω) is an almost Hermitian structure where ω is a symplectic form. One can show (see e.g. [1]) that the symplectic condition implies that

$$g((\nabla_Z^g J)X, Y) = -\frac{1}{2}g(N_J(X, Y), JZ), \quad (1)$$

where ∇^g denotes the Levi-Civita connection of g and N_J is the Nijenhuis tensor which we define with the following convention

$$N_J(X, Y) := J[JX, Y] + J[X, JY] + [X, Y] - [JX, JY].$$

Using equation (1), one can further show that

$$\nabla_{JX}^g J = -J\nabla_X^g J. \quad (2)$$

The Chern connection associated to (g, J, ω) is the unique connection D on TM for which $Dg = 0$, $DJ = 0$, and whose torsion T^D has vanishing $(1, 1)$

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part with respect to J . Explicitly, D is given by the following formula:

$$D_X Y = \nabla_X^g Y + \frac{1}{2}(\nabla_X^g J)(JY). \quad (3)$$

From this, it follows that the torsion is given by

$$T^D = -\frac{1}{4}N_J.$$

Let $T_J M$ be the complex vector bundle whose underlying real vector bundle is TM and where scalar multiplication by $\sqrt{-1}$ is given by

$$\sqrt{-1}X := JX.$$

As $DJ = 0$, we regard D as a connection on the complex vector bundle $T_J M$. Note that if $\dim M = 2n$, then $T_J M$ is a complex vector bundle of rank n . Let

$$\hat{g} := g - \sqrt{-1}\omega.$$

$\hat{g}$ then defines a natural hermitian metric on $T_J M$, that is,

$$\hat{g}(JX, Y) = \sqrt{-1}\hat{g}(X, Y), \quad \hat{g}(Y, X) = \overline{\hat{g}(X, Y)}.$$

Since g and J are both parallel with respect to D , it follows that $D\hat{g} = 0$. The curvature endomorphism of D is defined by

$$R^D(X, Y) := [D_X, D_Y] - D_{[X, Y]}.$$

The associated Chern-Ricci form is then

$$\rho_D(X, Y) := \text{Tr}(R^D(X, Y) \circ J) = \sqrt{-1}\text{Tr}(R^D(X, Y)), \quad (4)$$

where the second equality follows from the fact that $J = \sqrt{-1}\text{id}_{T_J M}$ under the identification $\sqrt{-1}X := JX$. In this paper, “Tr” will always mean the trace of a complex endomorphism. As a side remark, we recall that when the almost complex structure is integrable (i.e. the manifold is Kähler), the Chern-Ricci form coincides with the definition of the Ricci form.

Let

$$e_1, \dots, e_n, e_{n+1}, \dots, e_{2n}$$

be a g -orthonormal frame where $e_{n+i} := Je_i$ for $i \leq n$. Then $e_1, \dots, e_n$ is a unitary frame on $T_J M$ and we may write

$$D_X e_j = \sum_{i=1}^n A_{ij}(X) e_i$$

where A_{ij} are complex valued 1-forms whose domain is the same as the frame $\{e_i\}_{i=1}^n$. Taken together, $A := (A_{ij})$ is the local connection 1-form for D with respect to $\{e_i\}_{i=1}^n$.

The matrix representation of R^D with respect to $\{e_i\}_{i=1}^n$ is then the $n \times n$ matrix of 2-forms given by

$$[R^D] = dA + A \wedge A$$

where

$$(A \wedge A)(X, Y) := [A(X), A(Y)].$$

It follows from this and the cyclic properties of the trace that

$$\rho_D = \sqrt{-1} \, d\text{Tr}(A) = \sqrt{-1} \sum_{i=1}^n dA_{ii}. \quad (5)$$

Let $K^{-1} := \wedge_{\mathbb{C}}^n T_J M$ be the *anti-canonical bundle*. Then $\xi := e_1 \wedge \cdots \wedge e_n$ is a local section of K^{-1} . Applying D to K^{-1} one has

$$D_X \xi = \text{Tr}(A)(X) \cdot \xi.$$

Hence, $\text{Tr}(A)$ is the local connection 1-form of the pair (K^{-1}, D) . From this, it follows that the curvature of (K^{-1}, D) is

$$d\text{Tr}(A) = -\sqrt{-1} \rho_D.$$

Since $\widehat{g}(e_i, e_j) = \delta_{ij}$ and $D\widehat{g} = 0$, it follows that

$$A_{ji} = -\overline{A_{ij}}.$$

In particular, the diagonal elements A_{ii} are imaginary valued which, in turn, imply that the Chern-Ricci form ρ_D is real valued. As a consequence of this, equation (5) can be rewritten as

$$\rho_D = -d \, \text{Im} \, \text{Tr}(A). \quad (6)$$

For convenience, we make the following definition:

Definition 1. Let (M, g, J, ω) be an almost Kähler manifold. An automorphism $\psi \in \text{Aut}(TM)$ is called a *twist map* if it satisfies the following conditions:

- (i) ψ is g -symmetric, that is, ψ is self-adjoint with respect to g .
- (ii) $J\psi = \psi^{-1}J$.

On an almost Kähler manifold M , twist maps exist in abundance. Any smooth function $f \in C^\infty(M)$ gives rise to a twist map by the following construction¹ (see Theorem 4.7 of [2]). For $f \in C^\infty(M)$, let $X_f \in \mathfrak{X}(M)$ be the associated Hamiltonian vector field, that is,

$$\omega(X_f, \cdot) = -df.$$

Let $\widetilde{\psi} := \mathcal{L}_{X_f} J$ where $\mathcal{L}_{X_f}$ is the Lie derivative with respect to X_f . It follows immediately from this that

$$J\widetilde{\psi} = -\widetilde{\psi}J.$$

One can show that $\widetilde{\psi}$ is g -symmetric if and only if $\beta_f(Y, Z) = \beta_f(Z, Y)$ where

$$\beta_f(Y, Z) := g((\nabla_Y^g J)X_f, Z) - g(J(\nabla_{JY}^g J)X_f, Z)$$

However, by equation (2), it follows that $\beta_f \equiv 0$. Hence, $\widetilde{\psi}$ is g -symmetric. The g -symmetry of $\widetilde{\psi}$ and its anticommutativity with J implies that

$$\psi := \exp(\widetilde{\psi})$$

¹The authors wish to thank Professor Mehdi Lejmi for bringing this construction to their attention and for helpful discussions that motivated this work.

is a twist map. Using a twist map ψ , one obtains a new almost Kähler structure (g^ψ, J^ψ, ω) by defining

$$g^\psi := g(\psi^{-1}\cdot, \psi^{-1}\cdot), \quad J^\psi := \psi \circ J \circ \psi^{-1}. \quad (7)$$

Observe that the properties of ψ imply that

$$\omega^\psi := \omega(\psi^{-1}\cdot, \psi^{-1}\cdot) = \omega.$$

Hence, a twist map ψ deforms the metric and almost complex structure while leaving the Kähler form unchanged. We will call the new almost Kähler structure (g^ψ, J^ψ, ω) a ψ -twist of (g, J, ω) .

For convenience, let $(h, I, \omega) := (g^\psi, J^\psi, \omega)$. Also, let $\tilde{D}$ be the Chern connection associated to (M, h, I, ω) and let $e_1, \dots, e_n$ be a local $\hat{g}$ -unitary frame of $T_J M$. Define $\tilde{f}_1, \dots, \tilde{f}_n$ to be the $\hat{h}$ -unitary frame on $T_I M$ given by $\tilde{f}_i := \psi e_i$ and let $\tilde{A}$ be the local connection 1-form of $\tilde{D}$ with respect to $\tilde{f}_1, \dots, \tilde{f}_n$. In the current paper, we obtain the following formula² for $\text{Im Tr}(\tilde{A})$:

Theorem 2. *Let (M, g, J, ω) be an almost Kähler manifold and let ψ be a twist map. Let $h := g^\psi$ and $I := J^\psi$ and let $\tilde{D}$ be the Chern connection of (h, I, ω) . Let $e_1, \dots, e_n$ be a $\hat{g}$ -unitary frame of $T_J M$ and let*

$$\begin{aligned} \eta_P(X) := & -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi^{-1}[X, \psi e_i], e_i) \\ & - \frac{1}{2} \sum_{i=1}^{2n} g(J\psi \nabla_{\psi e_i}^g (\psi^{-2} X), e_i), \end{aligned} \quad (8)$$

where $e_{n+i} := J e_i$ for $i \leq n$. Let $\tilde{A}$ be the local connection 1-form of $\tilde{D}$ associated to the $\hat{h}$ -unitary frame $\tilde{f}_1, \dots, \tilde{f}_n$ on $T_I M$ where $\tilde{f}_i := \psi e_i$. Then $\text{Im Tr}(\tilde{A}) = \eta_P$ and $\rho_{\tilde{D}} = -d\eta_P$.

In addition, if

- (i) $(\psi e_i)g(e_j, J\psi e_i) = 0$ for all $i, j = 1, \dots, 2n$, and
- (ii) $[\psi e_i, \psi J e_i] = 0$ for $i \leq n$,

then

$$\eta_P(e_j) = -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi^{-1}[e_j, \psi e_i], e_i)$$

for $j = 1 \dots, 2n$.

The rest of the paper is organized as follows. In Section 2, we give a proof of Theorem 2. The paper concludes in Section 3 with some examples where we

²We note that P. Gauduchon derived an alternate formula (see equation (9.5.9) of [3]) for $\text{Im Tr}(\tilde{A})$. As far as we can tell, the two formulas do not seem to be derivable from one another by any simple algebraic transformation.

calculate $\text{Im Tr}(\tilde{A})$ using the formula η_P in Theorem 2. The examples considered satisfy conditions (i) and (ii) of Theorem 2 which further simplifies our calculation of η_P .

2. PROOF OF THEOREM 2

Let $e_1, \dots, e_n$ be a fixed $\hat{g}$ -unitary frame on $T_J M$. Then it follows that

$$e_1, \dots, e_n, e_{n+1}, \dots, e_{2n}$$

is a g -orthonormal frame on TM where $e_{n+i} := Je_i$ for $i \leq n$. Let $f_i := \psi e_i$ for $i \leq n$ and let $f_{n+i} := If_i = \psi e_{n+i}$ for $i \leq n$. Then

$$f_1, \dots, f_n, f_{n+1}, \dots, f_{2n}$$

is an h -orthonormal frame on TM . In particular, $f_1, \dots, f_n$ is an $\hat{h}$ -unitary frame on $T_I M$. Let $\tilde{A}$ be the connection 1-form of $\tilde{D}$ associated to $f_1, \dots, f_n$. From this, we have

$$\begin{aligned} \text{Tr}(\tilde{A})(X) &= \sum_{i=1}^n \hat{h}(\tilde{D}_X f_i, f_i) \\ &= -\sqrt{-1} \sum_{i=1}^n \omega(\tilde{D}_X f_i, f_i), \end{aligned}$$

where we use the fact that $\text{Tr}(\tilde{A})$ is purely imaginary since $f_1, \dots, f_n$ is an $\hat{h}$ -unitary frame. From this, we have

$$-\text{Im Tr}(\tilde{A})(X) = \sum_{i=1}^n \omega(\tilde{D}_X f_i, f_i). \quad (9)$$

Define

$$E(X, Y) := \nabla_X^h Y - \nabla_X^g Y, \quad (10)$$

where we recall that ∇^g and ∇^h are the Levi-Civita connections of g and h respectively. Using the almost Kähler identity (2), we can rewrite the Chern connection of (h, I, ω) as

$$\tilde{D}_X Y = \nabla_X^h Y + \frac{1}{2}(\nabla_{IX}^h I)Y. \quad (11)$$

Using (10), the second term in (11) expands as

$$\frac{1}{2}(\nabla_{IX}^h I)Y = \frac{1}{2}(\nabla_{IX}^g I)Y + \frac{1}{2}E(IX, IY) - \frac{1}{2}IE(IX, Y). \quad (12)$$

Rewriting (9) with the help of (11) and (12) gives

$$\begin{aligned}
-\operatorname{Im} \operatorname{Tr}(\tilde{A})(X) &= \sum_{i=1}^n \omega(\nabla_X^h f_i, f_i) + \frac{1}{2} \sum_{i=1}^n \omega((\nabla_{IX}^h I) f_i, f_i) \\
&= \sum_{i=1}^n \omega(\nabla_X^g f_i, f_i) + \sum_{i=1}^n \omega(E(X, f_i), f_i) \\
&\quad + \frac{1}{2} \sum_{i=1}^n \omega((\nabla_{IX}^g I) f_i, f_i) + \frac{1}{2} \sum_{i=1}^n \omega(E(IX, If_i), f_i) \\
&\quad - \frac{1}{2} \sum_{i=1}^n \omega(IE(IX, f_i), f_i). \tag{13}
\end{aligned}$$

Note that $If_1, \dots, If_n$ is also an $\hat{h}$ -unitary frame on $T_I M$. Moreover, the local connection 1-form of $\tilde{D}$ associated to the aforementioned frame is also $\tilde{A}$. Indeed, one has

$$\tilde{D}_X(If_j) = I\tilde{D}_X f_j = I \sum_{i=1}^n \tilde{A}_{ij}(X) f_i = \sum_{i=1}^n \tilde{A}_{ij}(X)(If_i).$$

Hence, (13) remains valid under the substitution $f_i \rightarrow If_i$:

$$\begin{aligned}
-\operatorname{Im} \operatorname{Tr}(\tilde{A})(X) &= \sum_{i=1}^n \omega(\nabla_X^g(If_i), If_i) + \sum_{i=1}^n \omega(E(X, If_i), If_i) \\
&\quad + \frac{1}{2} \sum_{i=1}^n \omega((\nabla_{IX}^g I)(If_i), If_i) - \frac{1}{2} \sum_{i=1}^n \omega(E(IX, f_i), If_i) \\
&\quad - \frac{1}{2} \sum_{i=1}^n \omega(IE(IX, If_i), If_i) \\
&= \sum_{i=1}^n \omega(\nabla_X^g(If_i), If_i) + \sum_{i=1}^n \omega(E(X, If_i), If_i) \\
&\quad - \frac{1}{2} \sum_{i=1}^n \omega((\nabla_{IX}^g I) f_i, f_i) + \frac{1}{2} \sum_{i=1}^n \omega(IE(IX, f_i), f_i) \\
&\quad - \frac{1}{2} \sum_{i=1}^n \omega(E(IX, If_i), f_i) \tag{14}
\end{aligned}$$

Taking the sum of (13) and (14) gives

$$\begin{aligned}
-2 \operatorname{Im} \operatorname{Tr}(\tilde{A})(X) &= \sum_{i=1}^n \omega(\nabla_X^g f_i, f_i) + \sum_{i=1}^n \omega(\nabla_X^g(If_i), If_i) \\
&\quad + \sum_{i=1}^n \omega(E(X, f_i), f_i) + \sum_{i=1}^n \omega(E(X, If_i), If_i). \tag{15}
\end{aligned}$$

For $i \leq n$, we now substitute $f_i = \psi e_i$ and $If_i = \psi(Je_i) = \psi e_{n+i}$ into (15) and rewrite all the ω 's in terms of g . This gives

$$\begin{aligned} -2 \operatorname{Im} \operatorname{Tr}(\tilde{A})(X) &= \sum_{i=1}^{2n} g(\psi J \nabla_X^g(\psi e_i), e_i) + \sum_{i=1}^{2n} g(\psi J E(X, \psi e_i), e_i) \\ &= \sum_{i=1}^{2n} g(J \psi^{-1} \nabla_X^g(\psi e_i), e_i) + \sum_{i=1}^{2n} g(J \psi^{-1} E(X, \psi e_i), e_i), \end{aligned} \quad (16)$$

where we have made use of the fact that ψ is g -symmetric and $\psi J = J \psi^{-1}$.

For convenience, we will often use the following notation:

$$\gamma_X := \nabla_X^g \psi^{-1} \quad (17)$$

Note that since ψ (and hence ψ^{-1}) is g -symmetric, one can easily show that γ_X is g -symmetric as well. To obtain the desired formula for $\operatorname{Im} \operatorname{Tr}(\tilde{A})$, we will need to compute E explicitly.

Proposition 3. *For $X, Y, Z \in \mathfrak{X}(M)$, let $Q(X, Y) \in \mathfrak{X}(M)$ be the unique vector field given by*

$$g(Q(X, Y), Z) = g(\psi^{-1} \gamma_Z X, Y) + g(\psi^{-1} \gamma_Z Y, X). \quad (18)$$

Then

$$E(X, Y) = \frac{1}{2} \psi [\gamma_X Y + \gamma_Y X + \psi \gamma_X \psi^{-1} Y + \psi \gamma_Y \psi^{-1} X - \psi Q(X, Y)].$$

Proof. By Proposition 2.1 of [4], we have

$$\begin{aligned} 2h(\nabla_X^h Y, Z) &= 2h(\nabla_X^g Y, Z) + h(\psi \gamma_X Y, Z) + h(\psi \gamma_Y X, Z) \\ &\quad + h(\psi \gamma_X Z, Y) - h(\psi \gamma_Z X, Y) + h(\psi \gamma_Y Z, X) \\ &\quad - h(\psi \gamma_Z Y, X). \end{aligned}$$

Applying the definition of h and using the g -symmetry of ψ^{-1} and γ_X , the above expression can be rewritten as

$$\begin{aligned} 2g(\psi^{-2} \nabla_X^h Y, Z) &= 2g(\psi^{-2} \nabla_X^g Y, Z) + g(\psi^{-1} \gamma_X Y, Z) + g(\psi^{-1} \gamma_Y X, Z) \\ &\quad + g(Z, \gamma_X \psi^{-1} Y) - g(\psi^{-1} \gamma_Z X, Y) + g(Z, \gamma_Y \psi^{-1} X) \\ &\quad - g(\psi^{-1} \gamma_Z Y, X). \end{aligned}$$

Applying the definition of Q , we have

$$\begin{aligned} 2g(\psi^{-2} \nabla_X^h Y, Z) &= 2g(\psi^{-2} \nabla_X^g Y, Z) + g(\psi^{-1} \gamma_X Y, Z) + g(\psi^{-1} \gamma_Y X, Z) \\ &\quad + g(\gamma_X \psi^{-1} Y, Z) + g(\gamma_Y \psi^{-1} X, Z) \\ &\quad - g(Q(X, Y), Z). \end{aligned}$$

The nondegeneracy of g implies that

$$\begin{aligned} 2\psi^{-2} \nabla_X^h Y &= 2\psi^{-2} \nabla_X^g Y + \psi^{-1} \gamma_X Y + \psi^{-1} \gamma_Y X \\ &\quad + \gamma_X \psi^{-1} Y + \gamma_Y \psi^{-1} X - Q(X, Y). \end{aligned}$$

Lastly, applying $\frac{1}{2}\psi^2$ to both sides gives

$$\nabla_X^h Y = \nabla_X^g Y + E(X, Y).$$

□

We now decompose the second term in (16) as follows:

$$\begin{aligned} \sum_{i=1}^{2n} g(J\psi^{-1}E(X, \psi e_i), e_i) &= \sum_{i=1}^n g(J\psi^{-1}E(X, \psi e_i), e_i) \\ &\quad + \sum_{i=1}^n g(J\psi^{-1}E(X, \psi J e_i), J e_i) \end{aligned} \quad (19)$$

where we recall that $e_{n+i} := J e_i$ for $i \leq n$. Using Proposition 3, the first term in (19) expands as

$$\begin{aligned} \sum_{i=1}^n g(J\psi^{-1}E(X, \psi e_i), e_i) &= - \sum_{i=1}^n g(\psi^{-1}E(X, \psi e_i), J e_i) \\ &= -\frac{1}{2} \sum_{i=1}^n g(\gamma_X(\psi e_i), J e_i) - \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi e_i} X, J e_i) \\ &\quad - \frac{1}{2} \sum_{i=1}^n g(\psi \gamma_X e_i, J e_i) - \frac{1}{2} \sum_{i=1}^n g(\psi \gamma_{\psi e_i}(\psi^{-1} X), J e_i) \\ &\quad + \frac{1}{2} \sum_{i=1}^n g(Q(X, \psi e_i), \psi J e_i) \\ &= -\frac{1}{2} \sum_{i=1}^n g(\psi \gamma_X(J e_i), e_i) - \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi e_i} X, J e_i) \\ &\quad - \frac{1}{2} \sum_{i=1}^n g(\gamma_X(\psi J e_i), e_i) - \frac{1}{2} \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi e_i}(\psi J e_i), X) \\ &\quad + \frac{1}{2} \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi J e_i} X, \psi e_i) + \frac{1}{2} \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi J e_i}(\psi e_i), X). \end{aligned} \quad (20)$$

The last two terms coming from the Q -term in (20) can be rewritten using g -symmetry. The result is then

$$\begin{aligned} \sum_{i=1}^n g(J\psi^{-1}E(X, \psi e_i), e_i) &= -\frac{1}{2} \sum_{i=1}^n g(\psi \gamma_X(J e_i), e_i) - \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi e_i} X, J e_i) \\ &\quad - \frac{1}{2} \sum_{i=1}^n g(\gamma_X(\psi J e_i), e_i) - \frac{1}{2} \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi e_i}(\psi J e_i), X) \\ &\quad + \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi J e_i} X, e_i) + \frac{1}{2} \sum_{i=1}^n g(\psi \gamma_{\psi J e_i}(\psi^{-1} X), e_i). \end{aligned} \quad (21)$$

In a completely similar way, we expand the second term in (19) using Proposition 3.

$$\begin{aligned}
\sum_{i=1}^n g(J\psi^{-1}E(X, \psi J e_i), J e_i) &= \sum_{i=1}^n g(\psi^{-1}E(X, \psi J e_i), e_i) \\
&= \frac{1}{2} \sum_{i=1}^n g(\gamma_X(\psi J e_i), e_i) + \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi J e_i} X, e_i) \\
&\quad + \frac{1}{2} \sum_{i=1}^n g(\psi \gamma_X(J e_i), e_i) + \frac{1}{2} \sum_{i=1}^n g(\psi \gamma_{\psi J e_i}(\psi^{-1} X), e_i) \\
&\quad - \frac{1}{2} \sum_{i=1}^n g(\gamma_{\psi e_i} X, J e_i) - \frac{1}{2} \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi e_i}(\psi J e_i), X). \tag{22}
\end{aligned}$$

Substituting (21) and (22) into (19) gives

$$\begin{aligned}
\sum_{i=1}^{2n} g(J\psi^{-1}E(X, \psi e_i), e_i) &= - \sum_{i=1}^n g(\gamma_{\psi e_i} X, J e_i) - \sum_{i=1}^n g(\psi^{-1} \gamma_{\psi e_i}(\psi J e_i), X) \\
&\quad + \sum_{i=1}^n g(\gamma_{\psi J e_i} X, e_i) + \sum_{i=1}^n g(\psi \gamma_{\psi J e_i}(\psi^{-1} X), e_i) \\
&= \sum_{i=1}^n g(J \gamma_{\psi e_i} X, e_i) + \sum_{i=1}^n g(J \psi \gamma_{\psi e_i}(\psi^{-1} X), e_i) \\
&\quad + \sum_{i=1}^n g(J \gamma_{\psi J e_i} X, J e_i) + \sum_{i=1}^n g(J \psi \gamma_{\psi J e_i}(\psi^{-1} X), J e_i) \tag{23}
\end{aligned}$$

Using $e_{n+i} := J e_i$ for $i \leq n$, (23) can be rewritten as

$$\sum_{i=1}^{2n} g(J\psi^{-1}E(X, \psi e_i), e_i) = \sum_{i=1}^{2n} g(J \gamma_{\psi e_i} X, e_i) + \sum_{i=1}^{2n} g(J \psi \gamma_{\psi e_i}(\psi^{-1} X), e_i). \tag{24}$$

Substituting (24) into (16) gives

$$\begin{aligned}
-2 \operatorname{Im} \operatorname{Tr}(\tilde{A})(X) &= \sum_{i=1}^{2n} g(J\psi^{-1} \nabla_X^g(\psi e_i), e_i) \\
&\quad + \sum_{i=1}^{2n} g(J(\nabla_{\psi e_i}^g \psi^{-1}) X, e_i) + \sum_{i=1}^{2n} g(J\psi(\nabla_{\psi e_i}^g \psi^{-1})(\psi^{-1} X), e_i), \tag{25}
\end{aligned}$$

where we have recalled that $\gamma_X := \nabla_X^g \psi^{-1}$. Define

$$\Xi_i := J\psi^{-1} \nabla_X^g(\psi e_i) + J(\nabla_{\psi e_i}^g \psi^{-1}) X + J\psi(\nabla_{\psi e_i}^g \psi^{-1})(\psi^{-1} X).$$

Expanding Ξ_i gives

$$\begin{aligned}
\Xi_i &= J\psi^{-1}\nabla_X^g(\psi e_i) + J\nabla_{\psi e_i}^g(\psi^{-1}X) \\
&\quad - J\psi^{-1}\nabla_{\psi e_i}^g X + J\psi\nabla_{\psi e_i}^g(\psi^{-2}X) \\
&\quad - J\nabla_{\psi e_i}^g(\psi^{-1}X) \\
&= J\psi^{-1}[X, \psi e_i] + J\psi\nabla_{\psi e_i}^g(\psi^{-2}X).
\end{aligned} \tag{26}$$

Applying (26) to (25) gives

$$\begin{aligned}
\text{Im Tr}(\tilde{A})(X) &= -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi^{-1}[X, \psi e_i], e_i) - \frac{1}{2} \sum_{i=1}^{2n} g(J\psi\nabla_{\psi e_i}^g(\psi^{-2}X), e_i) \\
&= \eta_P(X).
\end{aligned}$$

For the last statement of Theorem 2, let us now assume that conditions (i) and (ii) of Theorem 2 are satisfied. Let

$$S_1(X) := -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi^{-1}[X, \psi e_i], e_i)$$

and

$$S_2(X) := -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi\nabla_{\psi e_i}^g(\psi^{-2}X), e_i).$$

Then $\eta_P = S_1 + S_2$. Using condition (i), we observe that

$$\begin{aligned}
(\psi e_i)g(\psi^{-2}e_j, \psi J e_i) &= (\psi e_i)g(e_j, \psi^{-1}J e_i) \\
&= (\psi e_i)g(e_j, J\psi e_i) \\
&= 0.
\end{aligned}$$

The above identity together with the metric compatibility of ∇^g now implies

$$\begin{aligned}
S_2(e_j) &= -\frac{1}{2} \sum_{i=1}^{2n} g(J\psi\nabla_{\psi e_i}^g(\psi^{-2}e_j), e_i) \\
&= \frac{1}{2} \sum_{i=1}^{2n} g(\nabla_{\psi e_i}^g(\psi^{-2}e_j), \psi J e_i) \\
&= -\frac{1}{2} \sum_{i=1}^{2n} g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi J e_i)) \\
&= -\frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi J e_i)) - \frac{1}{2} \sum_{i=n+1}^{2n} g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi J e_i)) \\
&= -\frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi J e_i)) + \frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi J e_i}^g(\psi e_i))
\end{aligned}$$

where we have used the fact that $e_{n+i} := Je_i$ in the fifth equality. Using the fact that ∇^g is torsion free gives

$$\begin{aligned}
S_2(e_j) &= -\frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi Je_i)) + \frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi Je_i}^g(\psi e_i)) \\
&= -\frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi Je_i)) + \frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, \nabla_{\psi e_i}^g(\psi Je_i)) \\
&\quad + \frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, [\psi Je_i, \psi e_i]) \\
&= \frac{1}{2} \sum_{i=1}^n g(\psi^{-2}e_j, [\psi Je_i, \psi e_i]) \\
&= 0,
\end{aligned}$$

where the last equality follows from condition (ii) of Theorem 2. Hence, $\eta_P(e_j) = S_1(e_j)$ under the assumptions of conditions (i) and (ii) of Theorem 2.

3. EXAMPLES

In this section, we apply the new formula to a simple 4-dimensional (strictly) almost Kähler manifold. Let G be the Lie group $H \times \mathbb{R}$ where H is the 3-dimensional Heisenberg Lie group. Explicitly, H is given by

$$H = \left\{ \begin{pmatrix} 1 & x_1 & x_2 \\ 0 & 1 & x_3 \\ 0 & 0 & 1 \end{pmatrix} \mid x_1, x_2, x_3 \in \mathbb{R} \right\}.$$

G has global coordinates $x_1, \dots, x_4$ where x_4 is the natural coordinate on $\mathbb{R}$. Let $\mathfrak{h} := \text{Lie}(H)$ be the Lie algebra of left-invariant vector fields on H which we identify with the tangent space of H at the identity. A basis for $\mathfrak{h}$ is

$$\tilde{e}_1 := \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \tilde{e}_2 := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \tilde{e}_3 := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The only nonzero bracket on $\mathfrak{h}$ is $[\tilde{e}_1, \tilde{e}_2] = \tilde{e}_3$. The Lie algebra of G is $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{R}$. A basis for $\mathfrak{g}$ is

$$e_j := (\tilde{e}_j, 0), \quad j = 1, 2, 3, \quad e_4 := (0, 1).$$

Let $\partial_j := \frac{\partial}{\partial x_j}$ for $j = 1, \dots, 4$. Expressing the left-invariant vector fields $\{e_j\}$ in terms of the coordinate tangent vector fields one has

$$e_1 = \partial_1, \quad e_2 = x_1 \partial_2 + \partial_3, \quad e_3 = \partial_2, \quad e_4 = \partial_4.$$

Let $\theta^1, \dots, \theta^4$ denote the left-invariant 1-forms on $\mathfrak{g}$ dual to $e_1, \dots, e_4$. Then

$$dx_1 = \theta^1, \quad dx_2 = x_1 \theta^2 + \theta^3, \quad dx_3 = \theta^2, \quad dx_4 = \theta^4.$$

We consider a left-invariant almost Kähler structure (g, J, ω) on G which is defined as follows:

$$g(e_i, e_j) = \delta_{ij}, \quad \omega = \theta^1 \wedge \theta^3 + \theta^2 \wedge \theta^4,$$

$$Je_1 = e_3, \quad Je_2 = e_4, \quad Je_3 = -e_1, \quad Je_4 = -e_2.$$

We write the components of the Levi-Civita connection of g as $\nabla_{e_i}^g e_j = \sum_{k=1}^4 \Gamma_{ij}^k e_k$. The nonzero components are then

$$\Gamma_{23}^1 = \frac{1}{2}, \quad \Gamma_{32}^1 = \frac{1}{2}, \quad \Gamma_{13}^2 = -\frac{1}{2}, \quad \Gamma_{31}^2 = -\frac{1}{2}, \quad \Gamma_{12}^3 = \frac{1}{2}, \quad \Gamma_{21}^3 = -\frac{1}{2}.$$

Let S_1 and S_2 be defined as in Section 2. Then $\eta_P = S_1 + S_2$. Note that while η_P is a 1-form, S_1 and S_2 , considered individually, are not 1-forms. For the examples considered below, we choose functions f on G for which

$$\tilde{\psi} := \mathcal{L}_{X_f} J$$

is sufficiently simple, that is, the associated twist map $\psi := \exp(\tilde{\psi})$ can be computed explicitly. Moreover, the functions f have been chosen so that conditions (i) and (ii) of Theorem 2 are satisfied, which will further simplify the calculation of $\text{Im Tr}(\tilde{A})$.

Example 4. Let $f = \lambda x_2$ where $\lambda \in \mathbb{R}$. By direct calculation, one finds that

$$X_f = -\lambda e_1 + \lambda x_1 e_4.$$

$\tilde{\psi} := \mathcal{L}_{X_f} J$ is given by

$$\tilde{\psi} e_1 = -\lambda e_2, \quad \tilde{\psi} e_2 = -\lambda e_1, \quad \tilde{\psi} e_3 = \lambda e_4, \quad \tilde{\psi} e_4 = \lambda e_3.$$

$\psi := \exp(\tilde{\psi})$ is then given by

$$\begin{aligned} \psi e_1 &= ce_1 - se_2, & \psi e_2 &= -se_1 + ce_2, \\ \psi e_3 &= ce_3 + se_4, & \psi e_4 &= se_3 + ce_4, \end{aligned}$$

where $c := \cosh(\lambda)$ and $s := \sinh(\lambda)$. The inverse of ψ is easily computed:

$$\begin{aligned} \psi^{-1} e_1 &= ce_1 + se_2, & \psi^{-1} e_2 &= se_1 + ce_2 \\ \psi^{-1} e_3 &= ce_3 - se_4, & \psi^{-1} e_4 &= -se_3 + ce_4. \end{aligned}$$

Since g and ψ are left-invariant, it follows that

$$(\psi e_i)g(e_j, \psi e_i) = 0$$

for all i, j . Moreover, let $U := \text{span}\{e_1, e_2\}$ and $V := \text{span}\{e_3, e_4\}$. For this example, we have $\psi U \subset U$ and $\psi V \subset V$. Given that $[\mathfrak{g}, V] = 0$ and $JU \subset V$, we have

$$[\psi e_i, \psi J e_i] = 0$$

for $i = 1, 2$. By the second statement of Theorem 2, $S_2(e_j) = 0$ for all j . Hence,

$$\eta_P(e_j) = S_1(e_j) := -\frac{1}{2} \sum_{i=1}^4 g(J\psi^{-1}[e_j, \psi e_i], e_i).$$

In addition, since $[\mathfrak{g}, V] = 0$, we immediately have

$$\eta_P(e_3) = \eta_P(e_4) = 0.$$

Moreover, since $\psi e_3, \psi e_4 \in V := \text{span}\{e_3, e_4\}$, $\eta_P(e_j)$ further reduces to

$$\eta_P(e_j) = -\frac{1}{2} \sum_{i=1}^2 g(J\psi^{-1}[e_j, \psi e_i], e_i).$$

To determine $\eta_P(e_1)$, we compute

$$\begin{aligned} J\psi^{-1}[e_1, \psi e_1] &= cse_1 - s^2e_2 \\ J\psi^{-1}[e_1, \psi e_2] &= -c^2e_1 + cse_2. \end{aligned}$$

From this, it follows that

$$\eta_P(e_1) = -\frac{1}{2}cs - \frac{1}{2}cs = -cs.$$

To determine $\eta_P(e_2)$, we compute

$$\begin{aligned} J\psi^{-1}[e_2, \psi e_1] &= c^2e_1 - cse_2 \\ J\psi^{-1}[e_2, \psi e_2] &= -cse_1 + s^2e_2. \end{aligned}$$

This implies

$$\eta_P(e_2) = -\frac{1}{2}c^2 - \frac{1}{2}s^2 = -\frac{1}{2}(c^2 + s^2).$$

Putting everything together, we conclude that

$$\eta_P = -cs\theta^1 - \frac{1}{2}(c^2 + s^2)\theta^2,$$

where, again, $c := \cosh(\lambda)$ and $s := \sinh(\lambda)$.

For comparison, we also calculate the connection 1-form $\tilde{A}$ with respect to the twisted Chern connection $\tilde{D}$ on $T_I G$ with respect to the frame $f_1 := \psi e_1$, $f_2 := \psi e_2$. By a direct (and lengthy) calculation, the four components of $\tilde{A}$ are given by

$$\begin{aligned} \tilde{A}_{11} &= \sqrt{-1} \left(-\frac{cs}{2}\theta^1 - \frac{c^2}{2}\theta^2 \right) \\ \tilde{A}_{21} &= -\frac{c^2 + s^2}{4}\theta^3 + \frac{cs}{2}\theta^4 + \sqrt{-1} \left[\frac{c^2 + s^2}{4}\theta^1 + \frac{cs}{2}\theta^2 \right] \\ \tilde{A}_{12} &= \frac{c^2 + s^2}{4}\theta^3 - \frac{cs}{2}\theta^4 + \sqrt{-1} \left[\frac{c^2 + s^2}{4}\theta^1 + \frac{cs}{2}\theta^2 \right] \\ \tilde{A}_{22} &= \sqrt{-1} \left[-\frac{cs}{2}\theta^1 - \frac{s^2}{2}\theta^2 \right]. \end{aligned}$$

Hence,

$$\text{Im Tr}(\tilde{A}) = -cs\theta^1 - \frac{c^2 + s^2}{2}\theta^2 = \eta_P.$$

Example 5. Let $f = x_4^2$. From $\omega(X_f, \cdot) = -df$, we see that

$$X_f = -2x_4e_2.$$

Computing $\tilde{\psi} = \mathcal{L}_{X_f}J$, we obtain

$$\tilde{\psi}e_1 = 2x_4e_1, \quad \tilde{\psi}e_2 = 2e_2, \quad \tilde{\psi}e_3 = -2x_4e_3, \quad \tilde{\psi}e_4 = -2e_4.$$

Since $\tilde{\psi}$ is diagonal, $\psi := \exp(\tilde{\psi})$ is especially easy to calculate in the current example:

$$\psi e_1 = ae_1, \quad \psi e_2 = be_2, \quad \psi e_3 = a^{-1}e_3, \quad \psi e_4 = b^{-1}e_4,$$

where $a := e^{2x_4}$ and $b := e^2$. By inspection, one verifies that conditions (i) and (ii) of Theorem 2 are satisfied here. Hence, $\eta_P(e_j) = S_1(e_j)$ for $j = 1, \dots, 4$. By direct calculation, we have the following:

$$\begin{aligned} S_1(e_1) &= -\frac{1}{2} \sum_{i=1}^4 g(J\psi^{-1}[e_1, \psi e_i], e_i) = 0, \\ S_1(e_2) &= -\frac{1}{2} \sum_{i=1}^4 g(J\psi^{-1}[e_2, \psi e_i], e_i) = -\frac{1}{2}e^{4x_4}, \\ S_1(e_3) &= -\frac{1}{2} \sum_{i=1}^4 g(J\psi^{-1}[e_3, \psi e_i], e_i) = 0, \\ S_1(e_4) &= -\frac{1}{2} \sum_{i=1}^4 g(J\psi^{-1}[e_4, \psi e_i], e_i) = 0. \end{aligned}$$

From this, we see that

$$\eta_P = -\frac{1}{2}e^{4x_4}\theta^2.$$

For comparison, we also compute the local connection 1-form $\tilde{A}$ of $\tilde{D}$ with respect to $f_i := \psi e_i$, $i = 1, 2$:

$$\tilde{A} = \begin{pmatrix} -\frac{\sqrt{-1}}{2}e^{4x_4}\theta^2 & \sqrt{-1}\xi\theta^1 + \zeta\theta^3 \\ \sqrt{-1}\xi\theta^1 - \zeta\theta^3 & 0 \end{pmatrix},$$

where $\xi = \frac{1}{4}e^{2x_4+2} + e^{-2x_4-2}$ and $\zeta = \frac{1}{4}e^{6x_4+2} + e^{2x_4-2}$. From this, we have

$$\text{Im Tr}(\tilde{A}) = -\frac{1}{2}e^{4x_4}\theta^2$$

which is in agreement with η_P .

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